

Appendix: a learned Schrödinger-bridge coupling on the closure test

Question. Minibatch OT-CFM loses each jet's fine structure in the known-modification closure test ($T(J) = 0.8 [0.8J + 0.2K(J)]$, parent-disjoint unpaired pools): it pairs each jet with another real jet. Does an online bridge whose training endpoints are *generated by the model* do better? This tests the Schrödinger-bridge (SB) coupling of $\|x - y\|^2/2$ at noise ε in the model coordinates, not whether marginal matching finds the physical modification.

Online α -DSBM (De Bortoli et al. 2024, Alg. 1)

- one UVCAN-S net (32.8M) for both drifts; inputs: state, time, direction
- bridge $x_t = (1 - t)x_0 + tx_1 + \sqrt{\varepsilon t(1 - t)} z$; the same ε in the Euler-Maruyama sampler (30 steps); Appendix J endpoint preconditioning
- stage 1: bridge matching on independent real pairs; stage 2: EMA rollouts without gradient, real $X_0 \rightarrow \hat{X}_1$ trains the backward drift, real $X_1 \rightarrow \hat{X}_0$ the forward one
- no true pairs, OT, adversarial or shape losses, teacher
- $\varepsilon = 1$ fixed before scoring (midpoint noise 0.5 in z); $\varepsilon = 0.25$ run only because $\varepsilon = 1$ failed with its error dominated by sampling spread (pre-registered)

Arms (seed 0, one A6000 each, GPU hours incl. rollouts)

- **bridge** pretrained 0.5 h; α -DSBM: 0.5 + 1.5 h refinement; **bridge** continued: 0.5 + 1.5 h of independent-pair training (the matched control)
- **OT-CFM** with the same backbone: 2 h, exact minibatch OT, L^2
- $\dagger$ historical U-Net references (ODE): **paired** control, OT-CFM

Gaussian check first (same code, analytic SB answer): refinement reaches the SB coupling within 1-3% with a drift linear in x ; the paper's MLP overshoots 3-7%; the pretrained-only sampler (12% short) and a wrong-sign coupling fail.

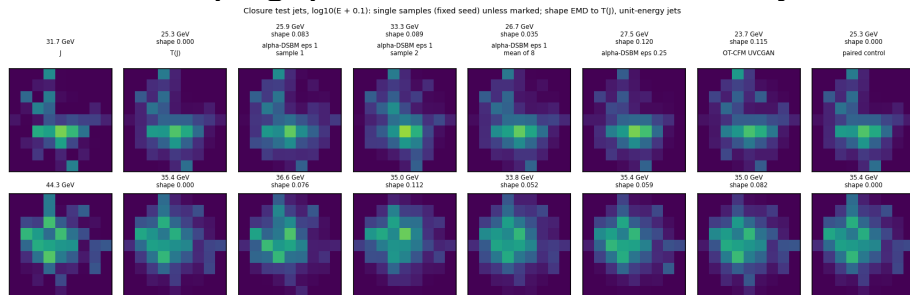
Result: better than not learning the coupling, still $2.4\text{-}3\times$ the identity

	shape EMD to $T(J)$				EMD	E_{out}/E_{in}	E RMSE	towers >5 GeV
	mean	median	p90	p99	GeV	(0.8)	GeV	mean/rms, GeV
identity $F(J) = J$	0.0322	0.0317	0.039	0.047	7.03	1.000 ± 0.000	6.61	4.05 / 4.45
paired control [†]	0.0002	0.0001	0.000	0.000	0.02	0.800 ± 0.000	0.02	-0.01 / 0.01
OT-CFM U-Net, L^2 [†]	0.0896	0.0768	0.151	0.274	3.56	0.785 ± 0.079	2.83	-1.82 / 3.44
OT-CFM U-Net, shape+ E [†]	0.0914	0.0825	0.143	0.240	2.87	0.808 ± 0.037	1.32	-0.87 / 2.16
OT-CFM U-Net, s+ E , pool 1024 [†]	0.0760	0.0695	0.116	0.188	2.37	0.802 ± 0.032	1.10	-0.50 / 1.61
OT-CFM UVCAN, L^2	0.0915	0.0791	0.152	0.284	4.02	0.778 ± 0.102	3.65	-1.49 / 3.60
bridge pretrained, $\varepsilon = 1$	0.1940	0.1507	0.360	0.731	7.62	0.799 ± 0.169	5.49	-2.97 / 5.36
bridge continued, $\varepsilon = 1$	0.1953	0.1520	0.363	0.729	7.56	0.788 ± 0.164	5.38	-3.11 / 5.35
α -DSBM, $\varepsilon = 1$	0.0966	0.0901	0.142	0.224	5.75	0.818 ± 0.170	5.49	-1.45 / 4.18
bridge pretrained, $\varepsilon = 0.25$	0.2053	0.1582	0.408	0.718	7.47	0.782 ± 0.153	5.12	-2.84 / 5.01
bridge continued, $\varepsilon = 0.25$	0.2172	0.1649	0.445	0.757	7.53	0.781 ± 0.145	4.82	-3.07 / 5.19
α -DSBM, $\varepsilon = 0.25$	0.0771	0.0678	0.124	0.237	4.42	0.796 ± 0.129	4.34	-0.44 / 3.38

20k held-out test pairs, one output per jet (bridges: one SDE sample, 30 steps, fixed seed; OT-CFM: ODE, 32 NFE); checkpoints selected on validation only. [†] historical U-Net runs (ODE).

Pre-registered gate: fails at both noise levels. $\varepsilon = 1$: not below OT-CFM with the same backbone (0.097 vs 0.092) and $3\times$ the identity. $\varepsilon = 0.25$: 16% below that OT-CFM and a better hard core (-0.44 vs -1.49 GeV), but $2.4\times$ the identity and worse distributions (mass $W1/\sigma$ 0.35 vs 0.12; mass change $1.38\times$ the truth, girth change $0.10\times$). **Measured:** at equal GPU hours the online refinement halves the error of continued independent-pair training (0.097 vs 0.195; 0.077 vs 0.217).

Why: at $\varepsilon = 1$ the sampling spread dominates; at $\varepsilon = 0.25$ a systematic error



	val, 1000 jets: shape EMD to $T(J)$			test, 1000 jets, 8 samples each			
	2 samples	30 vs 60 steps	one sample	2 samples	mean image	spread	E/σ_E
bridge pretrained, $\varepsilon = 1$	0.192	0.222	0.016	0.191	0.220	0.130	0.706
bridge continued, $\varepsilon = 1$	0.193	0.224	0.015	0.192	0.221	0.132	0.669
α -DSBM, $\varepsilon = 1$	0.099	0.116	0.013	0.096	0.118	0.057	0.700
bridge pretrained, $\varepsilon = 0.25$	0.201	0.195	0.012	0.200	0.191	0.158	0.531
bridge continued, $\varepsilon = 0.25$	0.211	0.209	0.015	0.212	0.204	0.167	0.507
α -DSBM, $\varepsilon = 0.25$	0.079	0.062	0.006	0.077	0.064	0.064	0.449

$\varepsilon = 1$: two samples of a jet differ more than either differs from $T(J)$; a jet's energy varies by 70% of the population spread. $\varepsilon = 0.25$: the spread halves and the mean of 8 samples is barely better than one: the error is systematic. 30 steps resolve the SDE.

Cost, and what this does and does not show

	GPU h	updates	updates/s	peak GB	selected at	NFE	ms / jet
paired control [†]	2.00	89.8k	12.5	2.9	2.00 h	32	2.97
OT-CFM U-Net, L^2 [†]	2.00	84.0k	11.7	3.0	1.33 h	32	2.96
OT-CFM U-Net, shape+ E [†]	2.00	81.5k	11.3	3.0	1.83 h	32	2.96
OT-CFM U-Net, s+ E , pool 1024 [†]	2.00	17.2k	2.4	3.0	2.00 h	32	2.96
OT-CFM UVCGAN, L^2	2.00	122.8k	17.1	3.5	1.67 h	32	1.62
bridge pretrained, ε 1	0.50	26.9k	14.9	3.6	0.50 h	30	1.53
bridge continued, ε 1	2.00	107.5k	14.9	3.8	1.67 h	30	1.53
α-DSBM , ε 1	2.00	35.9k	1.7	3.8	2.00 h	30	1.53
bridge pretrained, ε 0.25	0.50	27.0k	15.0	3.6	0.17 h	30	1.53
bridge continued, ε 0.25	2.00	108.2k	15.0	3.8	1.50 h	30	1.53
α-DSBM , ε 0.25	2.00	35.8k	1.6	3.8	2.00 h	30	1.53

One A6000 each. GPU h: pretraining and rollouts included; updates: all stages; updates/s: the last stage (**α -DSBM** spends 89% of its refinement in rollouts, 7680 NFE per update). Latency: batch 2000.

Measured

- the code reaches the analytic SB coupling when the network can represent it (Gaussian check)
- online refinement moves the coupling far from the independent-pair bridge at equal cost
- neither noise level approaches the identity's per-jet accuracy; $\varepsilon = 0.25$ was still improving at 2 GPU h (its limit is open, the gate is not: 0.077 vs 0.032)

Not shown

- algorithmic success is not the physical correspondence: the SB coupling of $\|x - y\|^2$ in $\log E$ is a choice, and T is not its answer
- robustness: one seed; no second seed or M $\rightarrow$ B pilot (gate failed)
- why $\varepsilon = 0.25$ distorts mass and girth (error amplification in the online loop, as for the Gaussian MLP, is a hypothesis)